

ADEN INTL, Yemen

WMO: 414800

Lat: 12.833N

Lon: 45.033E

Elev: 3

StdP: 101.29

Time Zone: 3.00 (E03)

Period: 02-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.2	22.0	11.2	8.3	28.1	13.5	9.6	27.3	10.9	26.5	10.3	26.3	2.2	30	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	6.1	37.0	24.8	36.2	25.0	35.9	25.1	28.9	33.2	28.3	32.9	28.0	32.6	5.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.8	23.9	32.1	27.1	22.9	32.0	26.8	22.4	31.9	94.2	33.3	91.7	33.1	89.9	32.3	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.8	9.5	8.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			13.8	29.7	1.6	0.9	12.7	30.4	11.7	30.9	10.8	31.5	9.7	32.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	29.2	25.8	25.8	26.6	28.8	31.3	32.6	32.2	32.0	29.3	27.3	26.0		
		DBStd	2.97	0.74	0.88	1.35	1.02	1.28	0.92	0.96	1.07	1.34	1.16	1.07		
		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0		
		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0		
		CDD10.0	7024	488	442	514	565	659	694	701	687	661	598	496		
		CDD18.3	3982	230	209	256	315	401	444	443	429	411	339	238		
		CDH23.3	51070	1723	1635	2493	3906	5756	6886	6673	6369	6051	4488	2964		
		CDH26.7	25362	142	177	566	1574	3283	4486	4193	3893	3653	2144	927		
(14)	Wind	WSAvg	4.7	5.8	5.7	5.4	5.4	4.1	3.6	4.3	4.5	3.9	4.4	4.7		
(15)	Precipitation	PrecAvg	60	9	3	8	3	3	1	5	6	4	9	5		
		PrecMax	158	97	11	32	42	9	10	12	58	27	99	51		
		PrecMin	19	0	0	0	0	1	0	2	0	1	0	2		
		PrecStd	37	20	3	9	9	2	2	3	16	6	23	10		
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.9	29.2	31.2	33.2	37.1	38.2	37.2	37.0	37.0	35.0	33.1		
			MCWB	21.4	22.3	22.5	24.6	24.9	25.7	24.8	24.3	24.6	24.0	21.2		
		2%	DB	28.2	28.5	30.2	32.2	35.9	37.1	36.2	36.1	36.0	33.7	31.8		
			MCWB	21.6	22.4	23.8	25.2	26.1	25.1	24.7	24.2	25.1	24.7	23.1		
		5%	DB	28.0	28.1	29.7	31.8	34.9	36.2	35.9	35.2	35.2	32.9	30.9		
			MCWB	21.7	22.2	23.6	25.2	26.2	25.3	24.8	24.7	25.6	25.1	23.6		
		10%	DB	27.4	27.9	29.1	31.1	33.9	35.9	35.1	35.0	34.8	32.1	30.0		
			MCWB	21.5	22.0	23.1	25.1	26.4	25.5	25.2	24.8	25.7	24.4	23.4		
(28)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.3	24.3	25.7	27.0	29.1	30.2	28.7	28.1	28.9	27.7	25.8		
			MCDB	27.2	28.2	29.5	31.1	32.9	32.9	32.7	32.4	33.5	32.3	30.1		
		2%	WB	22.8	23.7	25.2	26.5	28.4	29.2	27.9	27.4	28.3	27.1	25.1		
			MCDB	26.7	27.4	28.9	30.7	32.7	33.4	32.4	32.2	32.9	31.9	30.0		
		5%	WB	22.5	23.2	24.8	26.2	27.9	28.6	27.3	27.0	27.9	26.5	24.5		
			MCDB	26.5	26.9	28.5	30.3	32.5	33.3	32.3	32.0	32.5	31.5	29.4		
		10%	WB	22.2	22.7	24.2	25.9	27.4	28.1	26.9	26.6	27.4	26.0	24.1		
			MCDB	26.3	26.6	28.1	30.1	32.1	32.9	32.2	32.0	32.2	31.0	29.0		
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.2	4.5	5.3	4.9	5.2	6.1	5.8	6.1	5.7	6.3	5.9		
			MCDBR	4.3	4.6	5.5	5.5	6.8	6.7	6.4	6.4	6.6	6.9	6.8		
		5% WB	MCWBR	2.4	2.6	3.1	2.9	4.2	4.6	3.7	3.9	4.1	3.9	4.0		
			MCWBR	3.9	4.2	4.1	4.6	5.5	6.2	5.8	6.0	5.6	5.9	5.5		
(40)	Clear-Sky Solar Irradiance	taub	0.462	0.488	0.510	0.506	0.588	0.743	0.814	0.754	0.656	0.510	0.472	0.461		
		taud	2.177	2.089	2.044	2.083	1.881	1.569	1.481	1.582	1.735	2.047	2.152	2.177		
		Ebn at Noon	836	830	819	816	739	623	584	629	695	800	820	825		
		Edh at Noon	146	165	176	169	203	273	300	275	236	170	148	143		
(44)	All-Sky Solar Radiation	RadAvg	5.34	6.08	6.76	6.97	6.94	6.59	6.11	6.37	6.40	6.43	5.87	5.39		
		RadStd	0.23	0.25	0.25	0.25	0.23	0.25	0.22	0.17	0.16	0.18	0.17	0.25		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page